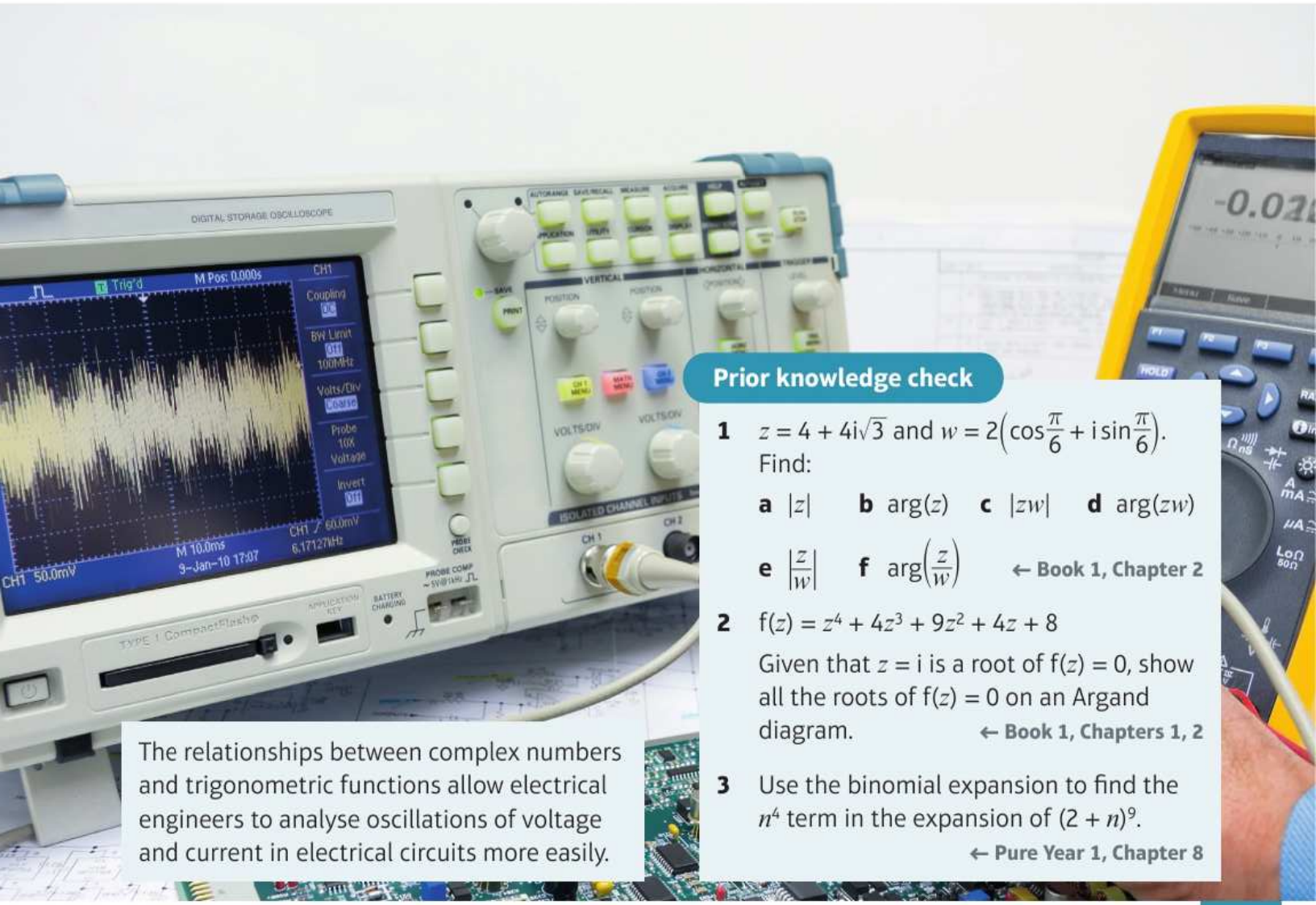
Complex numbers

1

Objectives

After completing this chapter you should be able to:

- Express a complex number in exponential form → pages 2–5
- Multiply and divide complex numbers in exponential form → pages 5–8
- Understand de Moivre's theorem → pages 8–11
- Use de Moivre's theorem to derive trigonometric identities → pages 11–15
- Use de Moivre's theorem to find sums of series → pages 16–19
- Know how to solve completely equations of the form $z^n - a - ib = 0$, giving special attention to cases where $a = 1$ and $b = 0$ → pages 20–25
- Use complex roots of unity to solve geometric problems → pages 25–27



The relationships between complex numbers and trigonometric functions allow electrical engineers to analyse oscillations of voltage and current in electrical circuits more easily.

Prior knowledge check

1 $z = 4 + 4i\sqrt{3}$ and $w = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$. Find:

a $|z|$ **b** $\arg(z)$ **c** $|zw|$ **d** $\arg(zw)$

e $\left|\frac{z}{w}\right|$ **f** $\arg\left(\frac{z}{w}\right)$ ← Book 1, Chapter 2

2 $f(z) = z^4 + 4z^3 + 9z^2 + 4z + 8$

Given that $z = i$ is a root of $f(z) = 0$, show all the roots of $f(z) = 0$ on an Argand diagram. ← Book 1, Chapters 1, 2

3 Use the binomial expansion to find the n^4 term in the expansion of $(2 + n)^9$.

← Pure Year 1, Chapter 8

1.1 Exponential form of complex numbers

You can use the modulus–argument form of a complex number to express it in the **exponential form**: $z = re^{i\theta}$.

Links The **modulus–argument** form of a complex number is $z = r(\cos\theta + i\sin\theta)$, where $r = |z|$ and $\theta = \arg z$.
← Book 1, Section 2.3

You can write $\cos\theta$ and $\sin\theta$ as infinite series of powers of θ:

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \ldots + \frac{(-1)^r\,\theta^{2r}}{(2r)!} + \ldots \quad (1)$$

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \ldots + \frac{(-1)^r\,\theta^{2r+1}}{(2r+1)!} + \ldots \quad (2)$$

You can also write e^x, $x \in \mathbb{R}$, as a series expansion in powers of x.

Links These are the Maclaurin series expansions of $\sin\theta$, $\cos\theta$ and e^x.
→ Chapter 2

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots + \frac{x^r}{r!} + \ldots$$

You can use this expansion to define the exponential function for complex powers, by replacing x with a complex number. In particular, if you replace x with the imaginary number $i\theta$, you get

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \ldots$$

$$= 1 + i\theta + \frac{i^2\theta^2}{2!} + \frac{i^3\theta^3}{3!} + \frac{i^4\theta^4}{4!} + \frac{i^5\theta^5}{5!} + \frac{i^6\theta^6}{6!} + \ldots$$

$$= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \frac{\theta^6}{6!} + \ldots$$

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \ldots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \ldots\right)$$

By comparing this series expansion with (1) and (2), you can write $e^{i\theta}$ as

$$e^{i\theta} = \cos\theta + i\sin\theta$$

This formula is known as **Euler's relation**. It is important for you to remember this result.

- **You can use Euler's relation, $e^{i\theta} = \cos\theta + i\sin\theta$, to write a complex number z in exponential form:**

  $$z = re^{i\theta}$$

  where $r = |z|$ and $\theta = \arg z$.

Note Substituting $\theta = \pi$ into Euler's relation yields **Euler's identity**:

$$e^{i\pi} + 1 = 0$$

This equation links the five fundamental constants 0, 1, π, e and i, and is considered an example of mathematical beauty.

Example 1

Express the following in the form $re^{i\theta}$, where $-\pi < \theta \leqslant \pi$.

a $z = \sqrt{2}\left(\cos\frac{\pi}{10} + i\sin\frac{\pi}{10}\right)$ **b** $z = 5\left(\cos\frac{\pi}{8} - i\sin\frac{\pi}{8}\right)$

a $z = \sqrt{2}\left(\cos\frac{\pi}{10} + i\sin\frac{\pi}{10}\right)$

Compare with $r(\cos\theta + i\sin\theta)$.

So $r = \sqrt{2}$ and $\theta = \frac{\pi}{10}$

Therefore, $z = \sqrt{2}e^{\frac{\pi i}{10}}$

$z = re^{i\theta}$

b $z = 5\left(\cos\frac{\pi}{8} - i\sin\frac{\pi}{8}\right)$

Problem-solving

Use $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.

$z = 5\left(\cos\left(-\frac{\pi}{8}\right) + i\sin\left(-\frac{\pi}{8}\right)\right)$

So $r = 5$ and $\theta = -\frac{\pi}{8}$

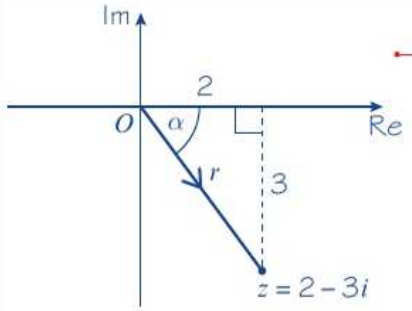
Compare with $r(\cos\theta + i\sin\theta)$.

Therefore, $z = 5e^{-\frac{\pi i}{8}}$

$z = re^{i\theta}$.

Example 2

Express $z = 2 - 3i$ in the form $re^{i\theta}$, where $-\pi < \theta \leqslant \pi$.



Sketch the Argand diagram, showing the position of the complex number.

Here z is in the fourth quadrant so the required argument is $-\alpha$.

$r = |z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

$\theta = \arg z = -\arctan\left(\frac{3}{2}\right) = -0.983$ (3 s.f.)

Find r and θ.

Therefore, $z = \sqrt{13}e^{-0.983i}$

$z = re^{i\theta}$.

Example 3

Express $z = \sqrt{2}e^{\frac{3\pi i}{4}}$ in the form $x + iy$, where $x, y \in \mathbb{R}$.

$z = \sqrt{2}e^{\frac{3\pi i}{4}}$, so $r = \sqrt{2}$ and $\theta = \frac{3\pi}{4}$ — Compare with $re^{i\theta}$.

$z = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$ — Write z in modulus–argument form.

$= \sqrt{2}\left(-\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right)$

Therefore, $z = -1 + i$ — Simplify.

Example 4

Express $z = 2e^{\frac{23\pi i}{5}}$ in the form $r(\cos\theta + i\sin\theta)$, where $-\pi < \theta \leqslant \pi$.

$z = 2e^{\frac{23\pi i}{5}}$, so $r = 2$ and $\theta = \frac{23\pi}{5}$ — Compare with $re^{i\theta}$.

$\frac{23\pi}{5} - 2\pi = \frac{13\pi}{5}$, $\frac{13\pi}{5} - 2\pi = \frac{3\pi}{5}$

$\frac{3\pi}{5}$ is in the range $-\pi < \theta \leqslant \pi$

So $z = 2\left(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}\right)$ — Write z in the form $r(\cos\theta + i\sin\theta)$.

Problem-solving

$\cos\theta = \cos(\theta + 2\pi)$ and $\sin\theta = \sin(\theta + 2\pi)$.

Subtract multiples of 2π from $\frac{23\pi}{5}$ until you find a value in the range $-\pi < \theta \leqslant \pi$.

Example 5

Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$.

$e^{i\theta} = \cos\theta + i\sin\theta$ (1)

$e^{-i\theta} = e^{i(-\theta)} = \cos(-\theta) + i\sin(-\theta)$

So $e^{-i\theta} = \cos\theta - i\sin\theta$ (2) — Use $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.

$e^{i\theta} + e^{-i\theta} = 2\cos\theta$ — Add (1) and (2).

$\Rightarrow \frac{e^{i\theta} + e^{-i\theta}}{2} = \cos\theta$ — Divide both sides by 2.

Hence, $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, as required.

Exercise 1A

1 Express the following in the form $re^{i\theta}$, where $-\pi < \theta \leqslant \pi$. Use exact values of r and θ where possible, or values to 3 significant figures otherwise.

a -3 **b** $6i$ **c** $-2\sqrt{3} - 2i$

d $-8 + i$ **e** $2 - 5i$ **f** $-2\sqrt{3} + 2i\sqrt{3}$

g $\sqrt{8}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ **h** $8\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$ **i** $2\left(\cos\frac{\pi}{5} - i\sin\frac{\pi}{5}\right)$

2 Express the following in the form $x + iy$ where $x, y \in \mathbb{R}$.

a $e^{\frac{\pi i}{3}}$ **b** $4e^{\pi i}$ **c** $3\sqrt{2}\,e^{\frac{\pi i}{4}}$

d $8e^{\frac{\pi i}{6}}$ **e** $3e^{-\frac{\pi i}{2}}$ **f** $e^{\frac{5\pi i}{6}}$

g $e^{-\pi i}$ **h** $3\sqrt{2}e^{-\frac{3\pi i}{4}}$ **i** $8e^{-\frac{4\pi i}{3}}$

3 Express the following in the form $r(\cos\theta + i\sin\theta)$, where $-\pi < \theta \leqslant \pi$.

a $e^{\frac{16\pi i}{13}}$ **b** $4e^{\frac{17\pi i}{5}}$ **c** $5e^{-\frac{9\pi i}{8}}$

(P) **4** Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$.

1.2 Multiplying and dividing complex numbers

You can apply the modulus–argument rules for multiplying and dividing complex numbers to numbers written in exponential form.

Recall that, for any two complex numbers z_1 and z_2,

- $|z_1 z_2| = |z_1||z_2|$
- $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

Links These results can be proved by considering the numbers z_1 and z_2 in the form $r(\cos\theta + i\sin\theta)$ and using the addition formulae for cos and sin. ← Book 1, Section 2.3

Applying these results to numbers in exponential form gives the following result:

■ **If $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$, then:**

- $z_1z_2 = r_1r_2e^{i(\theta_1 + \theta_2)}$
- $\frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1 - \theta_2)}$

Watch out You cannot automatically assume the laws of indices work the same way with complex numbers as with real numbers. This result only shows that they can be applied in these specific cases.

Example 6

a Express $2e^{\frac{\pi i}{6}} \times \sqrt{3}e^{\frac{\pi i}{3}}$ in the form $x + iy$.

b $z = 2 + 2i$, $\text{Im}(zw) = 0$ and $|zw| = 3|z|$
Use geometrical reasoning to find the two possibilities for w, giving your answers in exponential form.

a $2e^{\frac{\pi i}{6}} \times \sqrt{3}e^{\frac{\pi i}{3}} = (2 \times \sqrt{3})e^{i(\frac{\pi}{6} + \frac{\pi}{3})}$

$= 2\sqrt{3}e^{\frac{\pi i}{2}}$

$= 2\sqrt{3}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

$= 2\sqrt{3}(0 + i)$

$= 2i\sqrt{3}$

$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$

Simplify.

Convert the complex number to modulus–argument form.

b $|zw| = 3|z| \Rightarrow |w| = 3$

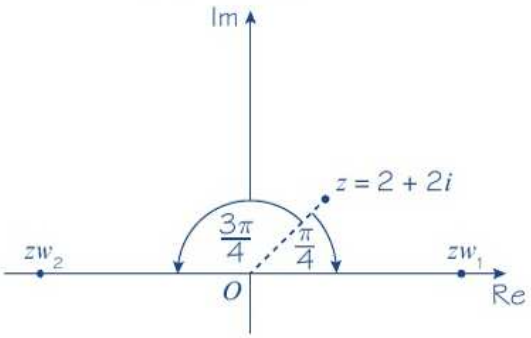
$\arg z = \arctan\left(\frac{2}{2}\right) = \frac{\pi}{4}$

$\text{Im}(zw) = 0$ so $\arg(zw) = 0$ or π

So $\arg w = \frac{3\pi}{4}$ or $-\frac{\pi}{4}$

$|zw| = |z||w| = 3|z|$.

wz lies on the real axis, so z is rotated $\frac{3\pi}{4}$ clockwise or $\frac{\pi}{4}$ anticlockwise when multiplied by w.



$w_1 = 3e^{-\frac{\pi i}{4}}$ and $w_2 = 3e^{\frac{3\pi i}{4}}$

Example 7

Express $\dfrac{2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)}{\sqrt{2}\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)}$ in the form $re^{i\theta}$.

$\dfrac{2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)}{\sqrt{2}\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)} = \dfrac{2e^{\frac{\pi i}{12}}}{\sqrt{2}e^{\frac{5\pi i}{6}}}$

$= \dfrac{2}{\sqrt{2}}e^{i(\frac{\pi}{12} - \frac{5\pi}{6})}$

$= \sqrt{2}e^{-\frac{3\pi i}{4}}$

Convert the numerator and denominator to exponential form.

$\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2}e^{i(\theta_1 - \theta_2)}$

Simplify.

Exercise 1B

1 Express the following in the form $x + iy$, where $x, y \in \mathbb{R}$.

a $e^{\frac{\pi i}{3}} \times e^{\frac{\pi i}{4}}$ **b** $\sqrt{5}e^{i\theta} \times 3e^{3i\theta}$ **c** $\sqrt{2}e^{\frac{2\pi i}{3}} \times e^{-\frac{7\pi i}{3}} \times 3e^{\frac{\pi i}{6}}$

2 Express the following in the form $x + iy$ where $x, y \in \mathbb{R}$.

a $\dfrac{2e^{\frac{7\pi i}{2}}}{8e^{\frac{9\pi i}{2}}}$ **b** $\dfrac{\sqrt{3}e^{\frac{3\pi i}{7}}}{4e^{-\frac{2\pi i}{7}}}$ **c** $\dfrac{\sqrt{2}e^{-\frac{15\pi i}{6}}}{2e^{\frac{\pi i}{3}}} \times \sqrt{2}\,e^{\frac{19\pi i}{3}}$

3 Express the following in the form $re^{i\theta}$.

a $(\cos 2\theta + i\sin 2\theta)(\cos 3\theta + i\sin 3\theta)$ **b** $\left(\cos\frac{3\pi}{11} + i\sin\frac{3\pi}{11}\right)\left(\cos\frac{8\pi}{11} + i\sin\frac{8\pi}{11}\right)$

c $3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \times 2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

d $\sqrt{6}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right) \times \sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

4 Express the following in the form $re^{i\theta}$.

a $\dfrac{\cos 5\theta + i\sin 5\theta}{\cos 2\theta + i\sin 2\theta}$ **b** $\dfrac{\sqrt{2}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)}{\frac{1}{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)}$ **c** $\dfrac{3\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)}{4\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)}$

5 z and w are two complex numbers where $z = -9 + 3i\sqrt{3}$, $|w| = \sqrt{3}$ and $\arg w = \dfrac{7\pi}{12}$

Express the following in the form $re^{i\theta}$, where $-\pi < \theta \leqslant \pi$.

a z **b** w **c** zw **d** $\dfrac{z}{w}$

(P) **6** Use the exponential form for a complex number to show that

$$\frac{(\cos 9\theta + i\sin 9\theta)(\cos 4\theta + i\sin 4\theta)}{\cos 7\theta + i\sin 7\theta} \equiv \cos 6\theta + i\sin 6\theta$$

(E/P) **7** $z = 1 + i\sqrt{3}$, $\mathrm{Re}\left(\dfrac{z^2}{w}\right) = 0$ and $\left|\dfrac{z^2}{w}\right| = |z|$

Use geometrical reasoning to find the two possibilities for w, giving your answers in exponential form. **(4 marks)**

(E/P) **8** **a** Evaluate $(1 + i)^2$, giving your answer in exponential form. **(2 marks)**

b Use mathematical induction to prove that $(1 + i)^n = 2^{\frac{n}{2}}e^{\frac{n\pi i}{4}}$ for $n \in \mathbb{Z}^+$. **(4 marks)**

c Hence find $(1 + i)^{16}$. **(1 mark)**

(P) **9** Use Euler's relation for $e^{i\theta}$ and $e^{-i\theta}$ to verify that $\cos^2\theta + \sin^2\theta \equiv 1$.

Challenge

a Given that n is a positive integer, prove by induction that $(re^{i\theta})^n = r^n e^{in\theta}$

b Given further that $z^{-n} = \frac{1}{z^n}$ for all $z \in \mathbb{C}$, show that $(re^{i\theta})^{-n} = r^{-n}e^{-in\theta}$

Watch out You cannot assume that the laws of indices will apply to complex numbers. Prove these results using only the properties

$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$

$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$

1.3 De Moivre's theorem

You can use Euler's relation to find powers of complex numbers given in modulus–argument form.

$$\begin{aligned}(r(\cos\theta + i\sin\theta))^2 &= (re^{i\theta})^2\\ &= re^{i\theta} \times re^{i\theta}\\ &= r^2 e^{i2\theta}\\ &= r^2(\cos 2\theta + i\sin 2\theta)\end{aligned}$$

Similarly, $(r(\cos\theta + i\sin\theta))^3 = r^3(\cos 3\theta + i\sin 3\theta)$, and so on.

The generalisation of this result is known as **de Moivre's theorem**:

- **For any integer n,**

  $(r(\cos\theta + i\sin\theta))^n = r^n(\cos n\theta + i\sin n\theta)$

You can prove de Moivre's theorem quickly using Euler's relation.

$$\begin{aligned}(r(\cos\theta + i\sin\theta))^n &= (re^{i\theta})^n\\ &= r^n e^{in\theta}\\ &= r^n(\cos n\theta + i\sin n\theta)\end{aligned}$$

This step is valid for any integer exponent n. ← Exercise 1B, Challenge

You can also prove de Moivre's theorem for **positive integer exponents** directly from the modulus–argument form of a complex number using the addition formulae for sin and cos.

Links This proof uses the method of proof by induction.

← Book 1, Chapter 8

1. Basis step

$n = 1$; LHS $= (r(\cos\theta + i\sin\theta))^1 = r(\cos\theta + i\sin\theta)$

RHS $= r^1(\cos 1\theta + i\sin 1\theta) = r(\cos\theta + i\sin\theta)$

As LHS = RHS, de Moivre's theorem is true for $n = 1$.

2. Assumption step

Assume that de Moivre's theorem is true for $n = k$, $k \in \mathbb{Z}^+$:

$(r(\cos\theta + i\sin\theta))^k = r^k(\cos k\theta + i\sin k\theta)$

3. Inductive step

When $n = k + 1$,

$$(r(\cos\theta + \mathrm{i}\sin\theta))^{k+1} = (r(\cos\theta + \mathrm{i}\sin\theta))^k \times r(\cos\theta + \mathrm{i}\sin\theta)$$
$$= r^k(\cos k\theta + \mathrm{i}\sin k\theta) \times r(\cos\theta + \mathrm{i}\sin\theta) \quad \text{By assumption step}$$
$$= r^{k+1}(\cos k\theta + \mathrm{i}\sin k\theta)(\cos\theta + \mathrm{i}\sin\theta)$$
$$= r^{k+1}((\cos k\theta\cos\theta - \sin k\theta\sin\theta) + \mathrm{i}(\sin k\theta\cos\theta + \cos k\theta\sin\theta))$$
$$= r^{k+1}(\cos(k\theta + \theta) + \mathrm{i}\sin(k\theta + \theta)) \quad \text{By addition formulae}$$
$$= r^{k+1}(\cos((k+1)\theta) + \mathrm{i}\sin((k+1)\theta))$$

Therefore, de Moivre's theorem is true when $n = k + 1$.

4. Conclusion step

If de Moivre's theorem is true for $n = k$, then it has been shown to be true for $n = k + 1$.

As de Moivre's theorem is true for $n = 1$, it is now proven to be true for all $n \in \mathbb{Z}^+$ by mathematical induction.

Links The corresponding proof for negative integer exponents is left as an exercise.

→ Exercise 1C, Challenge

Example 8

Simplify $\dfrac{\left(\cos\frac{9\pi}{17} + \mathrm{i}\sin\frac{9\pi}{17}\right)^5}{\left(\cos\frac{2\pi}{17} - \mathrm{i}\sin\frac{2\pi}{17}\right)^3}$

$$\frac{\left(\cos\frac{9\pi}{17} + \mathrm{i}\sin\frac{9\pi}{17}\right)^5}{\left(\cos\frac{2\pi}{17} - \mathrm{i}\sin\frac{2\pi}{17}\right)^3}$$
$$= \frac{\left(\cos\frac{9\pi}{17} + \mathrm{i}\sin\frac{9\pi}{17}\right)^5}{\left(\cos\left(-\frac{2\pi}{17}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{17}\right)\right)^3}$$
$$= \frac{\cos\frac{45\pi}{17} + \mathrm{i}\sin\frac{45\pi}{17}}{\cos\left(-\frac{6\pi}{17}\right) + \mathrm{i}\sin\left(-\frac{6\pi}{17}\right)}$$
$$= \cos\left(\frac{45\pi}{17} - \left(-\frac{6\pi}{17}\right)\right) + \mathrm{i}\sin\left(\frac{45\pi}{17} - \left(-\frac{6\pi}{17}\right)\right)$$
$$= \cos\frac{51\pi}{17} + \mathrm{i}\sin\frac{51\pi}{17}$$
$$= \cos 3\pi + \mathrm{i}\sin 3\pi$$
$$= \cos\pi + \mathrm{i}\sin\pi$$
$$= -1 + \mathrm{i}(0)$$

So $\dfrac{\left(\cos\frac{9\pi}{17} + \mathrm{i}\sin\frac{9\pi}{17}\right)^5}{\left(\cos\frac{2\pi}{17} - \mathrm{i}\sin\frac{2\pi}{17}\right)^3} = -1$

$\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$

Apply de Moivre's theorem to both the numerator and the denominator.

$\dfrac{z_1}{z_2} = \cos(\theta_1 - \theta_2) + \mathrm{i}\sin(\theta_1 - \theta_2)$

Simplify.

Subtract 2π from the argument.

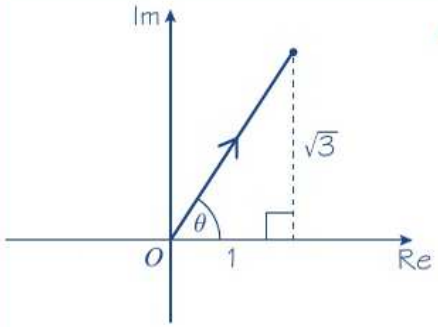
Problem-solving

You could also show this result by writing both numbers in exponential form:

$$\frac{\left(\mathrm{e}^{\frac{9\pi\mathrm{i}}{17}}\right)^5}{\left(\mathrm{e}^{-\frac{2\pi\mathrm{i}}{17}}\right)^3} = \frac{\mathrm{e}^{\frac{45\pi\mathrm{i}}{17}}}{\mathrm{e}^{-\frac{6\pi\mathrm{i}}{17}}} = \mathrm{e}^{\mathrm{i}\left(\frac{45\pi}{17} - \left(-\frac{6\pi}{17}\right)\right)} = \mathrm{e}^{3\pi\mathrm{i}} = \mathrm{e}^{\pi\mathrm{i}} = -1$$

Example 9

Express $(1 + i\sqrt{3})^7$ in the form $x + iy$ where $x, y \in \mathbb{R}$.



First, you need to find the modulus and argument of $1 + i\sqrt{3}$. You may want to draw an Argand diagram to help you.

$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$

$\theta = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$

Find r and θ.

So $1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

Write $1 + i\sqrt{3}$ in modulus–argument form.

$(1 + i\sqrt{3})^7 = \left(2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\right)^7$

$= 2^7\left(\cos\frac{7\pi}{3} + i\sin\frac{7\pi}{3}\right)$

Apply de Moivre's theorem.

$= 128\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

Subtract 2π from the argument.

$= 128\left(\frac{1}{2} + i\left(\frac{\sqrt{3}}{2}\right)\right)$

Therefore, $(1 + i\sqrt{3})^7 = 64 + 64i\sqrt{3}$

Exercise 1C

1 Use de Moivre's theorem to express each of the following in the form $x + iy$, where $x, y \in \mathbb{R}$.

a $(\cos\theta + i\sin\theta)^6$ **b** $(\cos 3\theta + i\sin 3\theta)^4$ **c** $\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)^5$

d $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^8$ **e** $\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)^5$ **f** $\left(\cos\frac{\pi}{10} - i\sin\frac{\pi}{10}\right)^{15}$

2 Express each of the following in the form $e^{in\theta}$.

a $\dfrac{\cos 5\theta + i\sin 5\theta}{(\cos 2\theta + i\sin 2\theta)^2}$ **b** $\dfrac{(\cos 2\theta + i\sin 2\theta)^7}{(\cos 4\theta + i\sin 4\theta)^3}$ **c** $\dfrac{1}{(\cos 2\theta + i\sin 2\theta)^3}$

d $\dfrac{(\cos 2\theta + i\sin 2\theta)^4}{(\cos 3\theta + i\sin 3\theta)^3}$ **e** $\dfrac{\cos 5\theta + i\sin 5\theta}{(\cos 3\theta - i\sin 3\theta)^2}$ **f** $\dfrac{\cos\theta - i\sin\theta}{(\cos 2\theta - i\sin 2\theta)^3}$

3 Evaluate the following, giving your answers in the form $x + iy$, where $x, y \in \mathbb{R}$.

a $\dfrac{\left(\cos\frac{7\pi}{13} - i\sin\frac{7\pi}{13}\right)^4}{\left(\cos\frac{4\pi}{13} + i\sin\frac{4\pi}{13}\right)^6}$ **b** $\dfrac{\left(\cos\frac{3\pi}{7} - i\sin\frac{11\pi}{7}\right)^3}{\left(\cos\frac{15\pi}{7} + i\sin\frac{\pi}{7}\right)^2}$ **c** $\dfrac{\left(\cos\frac{4\pi}{3} - i\sin\frac{2\pi}{3}\right)^7}{\left(\cos\frac{10\pi}{3} + i\sin\frac{4\pi}{3}\right)^4}$

4 Express the following in the form $x + iy$ where $x, y \in \mathbb{R}$.

a $(1 + i)^5$ **b** $(-2 + 2i)^8$ **c** $(1 - i)^6$

d $(1 - i\sqrt{3})^6$ **e** $\left(\frac{3}{2} - \frac{1}{2}i\sqrt{3}\right)^9$ **f** $(-2\sqrt{3} - 2i)^5$

E **5** Express $(3 + i\sqrt{3})^5$ in the form $a + bi\sqrt{3}$ where a and b are integers. **(2 marks)**

E **6** $w = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

Find the exact value of w^4, giving your answer in the form $a + ib$ where $a, b \in \mathbb{R}$. **(2 marks)**

E **7** $z = \sqrt{3}\left(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4}\right)$

Find the exact value of z^6, giving your answer in the form $a + ib$ where $a, b \in \mathbb{R}$. **(3 marks)**

E/P **8 a** Express $\dfrac{1 + i\sqrt{3}}{1 - i\sqrt{3}}$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leqslant \pi$. **(3 marks)**

b Hence find the smallest positive integer value of n for which $\left(\dfrac{1 + i\sqrt{3}}{1 - i\sqrt{3}}\right)^n$ is real and positive. **(2 marks)**

E/P **9** Use de Moivre's theorem to show that $(a + bi)^n + (a - bi)^n$ is real for all integers n. **(5 marks)**

Challenge

Without using Euler's relation, prove that if n is a positive integer,

$(r(\cos\theta + i\sin\theta))^{-n} = r^{-n}(\cos(-n\theta) + i\sin(-n\theta))$

Problem-solving

You may assume de Moivre's theorem for positive integer exponents, but do not write any complex numbers in exponential form.

1.4 Trigonometric identities

You can use de Moivre's theorem to derive trigonometric identities.

Applying the binomial expansion to $(\cos\theta + i\sin\theta)^n$ allows you to express $\cos n\theta$ in terms of powers of $\cos\theta$, and $\sin n\theta$ in terms of powers of $\sin\theta$.

Links $(a + b)^n = a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + \ldots + {}^nC_r a^{n-r}b^r + \ldots + b^n, n \in \mathbb{N}$

where ${}^nC_r = \binom{n}{r} = \dfrac{n!}{r!(n-r)!}$ ← Pure Year 1, Chapter 8

Example 10

Use de Moivre's theorem to show that

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

$(\cos\theta + \mathrm{i}\sin\theta)^6 = \cos 6\theta + \mathrm{i}\sin 6\theta$ — Apply de Moivre's theorem.

$= \cos^6\theta + {}^6C_1\cos^5\theta(\mathrm{i}\sin\theta) + {}^6C_2\cos^4\theta(\mathrm{i}\sin\theta)^2$
$+ {}^6C_3\cos^3\theta(\mathrm{i}\sin\theta)^3 + {}^6C_4\cos^2\theta(\mathrm{i}\sin\theta)^4$
$+ {}^6C_5\cos\theta(\mathrm{i}\sin\theta)^5 + (\mathrm{i}\sin\theta)^6$ — Apply the binomial expansion to $(\cos\theta + \mathrm{i}\sin\theta)^6$.

$= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta + 15\mathrm{i}^2\cos^4\theta\sin^2\theta$
$+ 20\mathrm{i}^3\cos^3\theta\sin^3\theta + 15\mathrm{i}^4\cos^2\theta\sin^4\theta$
$+ 6\mathrm{i}^5\cos\theta\sin^5\theta + \mathrm{i}^6\sin^6\theta$ — Simplify.

$= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta$
$- 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta$
$+ 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta$ — Simplify the powers of i.

Equating the real parts gives — The real part of $\cos 6\theta + \mathrm{i}\sin 6\theta$ is $\cos 6\theta$.

$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta$
$+ 15\cos^2\theta\sin^4\theta - \sin^6\theta$

$= \cos^6\theta - 15\cos^4\theta(1 - \cos^2\theta)$
$+ 15\cos^2\theta(1 - \cos^2\theta)^2 - (1 - \cos^2\theta)^3$ — Apply $\sin^2\theta \equiv 1 - \cos^2\theta$, $\sin^4\theta \equiv (\sin^2\theta)^2$ and $\sin^6\theta \equiv (\sin^2\theta)^3$.

$= \cos^6\theta - 15\cos^4\theta(1 - \cos^2\theta)$
$+ 15\cos^2\theta(1 - 2\cos^2\theta + \cos^4\theta)$ — Multiply out the brackets.
$- (1 - 3\cos^2\theta + 3\cos^4\theta - \cos^6\theta)$ — Apply a cubic binomial expansion.

$= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta$
$+ 15\cos^2\theta - 30\cos^4\theta + 15\cos^6\theta$
$- 1 + 3\cos^2\theta - 3\cos^4\theta + \cos^6\theta$ — Expand the brackets.

$= 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$ — Simplify.

Therefore,

$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$

You can also find trigonometric identities for $\sin^n\theta$ and $\cos^n\theta$ where n is a positive integer.

If $z = \cos\theta + \mathrm{i}\sin\theta$, then

$$\frac{1}{z} = z^{-1} = (\cos\theta + \mathrm{i}\sin\theta)^{-1}$$
$$= (\cos(-\theta) + \mathrm{i}\sin(-\theta)) \quad \text{Apply de Moivre's theorem.}$$
$$= \cos\theta - \mathrm{i}\sin\theta \quad \text{Use } \cos\theta = \cos(-\theta) \text{ and } -\sin\theta = \sin(-\theta).$$

It follows that

$$z + \frac{1}{z} = \cos\theta + \mathrm{i}\sin\theta + \cos\theta - \mathrm{i}\sin\theta = 2\cos\theta$$

$$z - \frac{1}{z} = \cos\theta + \mathrm{i}\sin\theta - (\cos\theta - \mathrm{i}\sin\theta) = 2\mathrm{i}\sin\theta$$

Also,

$z^n = (\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ — By de Moivre's theorem.

$\frac{1}{z^n} = z^{-n} = (\cos\theta + \mathrm{i}\sin\theta)^{-n}$

$= (\cos(-n\theta) + \mathrm{i}\sin(-n\theta))$ — Apply de Moivre's theorem.

$= \cos n\theta - \mathrm{i}\sin n\theta$ — Use $\cos\theta = \cos(-\theta)$ and $\sin(-\theta) = -\sin\theta$.

It follows that

$$z^n + \frac{1}{z^n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$$

$$z^n - \frac{1}{z^n} = \cos n\theta + \mathrm{i}\sin n\theta - (\cos n\theta - \mathrm{i}\sin n\theta) = 2\mathrm{i}\sin n\theta$$

It is important that you remember and are able to apply these results:

- $\boldsymbol{z + \frac{1}{z} = 2\cos\theta}$
- $\boldsymbol{z^n + \frac{1}{z^n} = 2\cos n\theta}$
- $\boldsymbol{z - \frac{1}{z} = 2\mathrm{i}\sin\theta}$
- $\boldsymbol{z^n - \frac{1}{z^n} = 2\mathrm{i}\sin n\theta}$

Notation In exponential form, these results are equivalent to:

$\cos n\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}n\theta} + \mathrm{e}^{-\mathrm{i}n\theta})$ $\quad$ $\sin n\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}n\theta} - \mathrm{e}^{-\mathrm{i}n\theta})$.

Example 11

Express $\cos^5\theta$ in the form $a\cos 5\theta + b\cos 3\theta + c\cos\theta$, where a, b and c are constants.

Let $z = \cos\theta + \mathrm{i}\sin\theta$

$\left(z + \frac{1}{z}\right)^5 = (2\cos\theta)^5 = 32\cos^5\theta$ — Use $z + \frac{1}{z} = 2\cos\theta$.

$= z^5 + {}^5C_1 z^4\left(\frac{1}{z}\right) + {}^5C_2 z^3\left(\frac{1}{z}\right)^2 + {}^5C_3 z^2\left(\frac{1}{z}\right)^3$

$+ {}^5C_4 z\left(\frac{1}{z}\right)^4 + \left(\frac{1}{z}\right)^5$ — Apply the binomial expansion to $\left(z + \frac{1}{z}\right)^5$.

$= z^5 + 5z^4\left(\frac{1}{z}\right) + 10z^3\left(\frac{1}{z^2}\right) + 10z^2\left(\frac{1}{z^3}\right)$

$+ 5z\left(\frac{1}{z^4}\right) + \left(\frac{1}{z^5}\right)$ — Simplify.

$= z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}$

$= \left(z^5 + \frac{1}{z^5}\right) + 5\left(z^3 + \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right)$ — Group z^n and $\frac{1}{z^n}$ terms.

$= 2\cos 5\theta + 5(2\cos 3\theta) + 10(2\cos\theta)$ — Use $z^n + \frac{1}{z^n} = 2\cos n\theta$.

So, $32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$

$\Rightarrow \quad \cos^5\theta = \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta$ — This is in the required form with $a = \frac{1}{16}$, $b = \frac{5}{16}$ and $c = \frac{5}{8}$

Example 12

a Express $\sin^4\theta$ in the form $d\cos 4\theta + e\cos 2\theta + f$, where d, e and f are constants.

b Hence find the exact value of $\int_0^{\frac{\pi}{2}} \sin^4\theta \, d\theta$.

a Let $z = \cos\theta + i\sin\theta$

$\left(z - \frac{1}{z}\right)^4 = (2i\sin\theta)^4 = 16i^4\sin^4\theta = 16\sin^4\theta$

Use $z - \frac{1}{z} = 2i\sin\theta$, noting that $i^4 = 1$

$= z^4 + {}^4C_1 z^3\left(-\frac{1}{z}\right) + {}^4C_2 z^2\left(-\frac{1}{z}\right)^2 + {}^4C_3 z^1\left(-\frac{1}{z}\right)^3 + \left(-\frac{1}{z}\right)^4$

Apply the binomial expansion to $\left(z - \frac{1}{z}\right)^4$

$= z^4 + 4z^3\left(-\frac{1}{z}\right) + 6z^2\left(\frac{1}{z^2}\right) + 4z\left(-\frac{1}{z^3}\right) + \left(\frac{1}{z^4}\right)$

$= z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4}$

Simplify.

$= \left(z^4 + \frac{1}{z^4}\right) - 4\left(z^2 + \frac{1}{z^2}\right) + 6$

Group z^n and $\frac{1}{z^n}$ terms.

$= 2\cos 4\theta - 4(2\cos 2\theta) + 6$

Use $z^n + \frac{1}{z^n} = 2\cos n\theta$.

So, $16\sin^4\theta = 2\cos 4\theta - 8\cos 2\theta + 6$

$\Rightarrow \quad \sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8}$

This is in the required form with $d = \frac{1}{8}$, $e = -\frac{1}{2}$ and $f = \frac{3}{8}$

b $\int_0^{\frac{\pi}{2}} \sin^4\theta \, d\theta = \int_0^{\frac{\pi}{2}} \left(\frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8}\right) d\theta$

Use the answer from part **a**.

$= \left[\frac{1}{32}\sin 4\theta - \frac{1}{4}\sin 2\theta + \frac{3}{8}\theta\right]_0^{\frac{\pi}{2}}$

$= \left(\frac{1}{32}\sin 2\pi - \frac{1}{4}\sin\pi + \frac{3}{8}\left(\frac{\pi}{2}\right)\right) - 0$

$= 0 - 0 + \frac{3\pi}{16}$

$= \frac{3\pi}{16}$

$\cos k\theta$ integrates to $\frac{1}{k}\sin k\theta$.

Exercise 1D

Use de Moivre's theorem to prove the following trigonometric identities:

(P) **1** **a** $\sin 3\theta \equiv 3\sin\theta - 4\sin^3\theta$

b $\sin 5\theta \equiv 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$

c $\cos 7\theta \equiv 64\cos^7\theta - 112\cos^5\theta + 56\cos^3\theta - 7\cos\theta$

d $\cos^4\theta \equiv \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$

e $\sin^5\theta \equiv \frac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$

E/P **2** **a** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **(5 marks)**

b Hence, given also that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, find all the solutions of $\cos 5\theta + 5\cos 3\theta = 0$ in the interval $0 \leqslant \theta < \pi$. Give your answers to 3 decimal places. **(6 marks)**

E/P **3** **a** Show that $32\cos^6\theta \equiv \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10$. **(6 marks)**

b Hence find $\int_0^{\frac{\pi}{6}} \cos^6\theta \, d\theta$ in the form $a\pi + b\sqrt{3}$ where a and b are rational constants to be found. **(3 marks)**

E/P **4** **a** Show that $32\cos^2\theta\sin^4\theta \equiv \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2$. **(6 marks)**

b Hence find the exact value of $\int_0^{\frac{\pi}{3}} \cos^2\theta\sin^4\theta \, d\theta$. **(3 marks)**

P **5** By using de Moivre's theorem, or otherwise, compute the following integrals.

a $\int_0^{\frac{\pi}{2}} \sin^6\theta \, d\theta$ **b** $\int_0^{\frac{\pi}{4}} \sin^2\theta\cos^4\theta \, d\theta$ **c** $\int_0^{\frac{\pi}{6}} \sin^3\theta\cos^5\theta \, d\theta$

E/P **6** **a** Use de Moivre's theorem to show that

$$\cos 6\theta \equiv 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$ **(5 marks)**

b Hence find the six distinct solutions of the equation

$$32x^6 - 48x^4 + 18x^2 - \tfrac{3}{2} = 0$$

giving your answers to 3 decimal places where necessary. **(5 marks)**

Problem-solving

Use the substitution $x = \cos\theta$ to reduce the equation to the form $\cos 6\theta = k$. Find as many values of θ as you need to find six distinct values of x.

E/P **7** **a** Use de Moivre's theorem to show that $\sin 4\theta \equiv 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$. **(4 marks)**

b Hence, or otherwise, show that $\tan 4\theta \equiv \dfrac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$ **(4 marks)**

c Use your answer to part **b** to find, to 2 decimal places, the four solutions of the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. **(5 marks)**

E/P **2** **a** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **(5 marks)**

b Hence, given also that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, find all the solutions of $\cos 5\theta + 5\cos 3\theta = 0$ in the interval $0 \leqslant \theta < \pi$. Give your answers to 3 decimal places. **(6 marks)**

E/P **3** **a** Show that $32\cos^6\theta \equiv \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10$. **(6 marks)**

b Hence find $\int_0^{\frac{\pi}{6}} \cos^6\theta \, d\theta$ in the form $a\pi + b\sqrt{3}$ where a and b are rational constants to be found. **(3 marks)**

E/P **4** **a** Show that $32\cos^2\theta\sin^4\theta \equiv \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2$. **(6 marks)**

b Hence find the exact value of $\int_0^{\frac{\pi}{3}} \cos^2\theta\sin^4\theta \, d\theta$. **(3 marks)**

P **5** By using de Moivre's theorem, or otherwise, compute the following integrals.

a $\int_0^{\frac{\pi}{2}} \sin^6\theta \, d\theta$ **b** $\int_0^{\frac{\pi}{4}} \sin^2\theta\cos^4\theta \, d\theta$ **c** $\int_0^{\frac{\pi}{6}} \sin^3\theta\cos^5\theta \, d\theta$

E/P **6** **a** Use de Moivre's theorem to show that

$$\cos 6\theta \equiv 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$ **(5 marks)**

b Hence find the six distinct solutions of the equation

$$32x^6 - 48x^4 + 18x^2 - \tfrac{3}{2} = 0$$

giving your answers to 3 decimal places where necessary. **(5 marks)**

Problem-solving

Use the substitution $x = \cos\theta$ to reduce the equation to the form $\cos 6\theta = k$. Find as many values of θ as you need to find six distinct values of x.

E/P **7** **a** Use de Moivre's theorem to show that $\sin 4\theta \equiv 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$. **(4 marks)**

b Hence, or otherwise, show that $\tan 4\theta \equiv \dfrac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$ **(4 marks)**

c Use your answer to part **b** to find, to 2 decimal places, the four solutions of the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. **(5 marks)**

1.5 Sums of series

You can use results about the sums of geometric series with complex numbers.

- **For $w, z \in \mathbb{C}$,**
  - $\sum_{r=0}^{n-1} wz^r = w + wz + wz^2 + \ldots + wz^{n-1} = \frac{w(z^n - 1)}{z - 1}$
  - $\sum_{r=0}^{\infty} wz^r = w + wz + wz^2 + \ldots = \frac{w}{1 - z}, |z| < 1$

Links These results match the corresponding results for real numbers. The infinite series $\sum_{r=0}^{\infty} wz^r$ converges only when $|z| < 1$.

← Pure Year 2, Chapter 3

Example 13

Given that $z = \cos\frac{\pi}{n} + \mathrm{i}\sin\frac{\pi}{n}$, where n is a positive integer, show that

$$1 + z + z^2 + \ldots + z^{n-1} = 1 + \mathrm{i}\cot\left(\frac{\pi}{2n}\right)$$

$$1 + z + z^2 + \ldots + z^{n-1} = \frac{z^n - 1}{z - 1}$$

Use the result for $\sum_{r=0}^{n-1} wz^r$ with $w = 1$.

$$= \frac{\left(e^{\frac{\pi \mathrm{i}}{n}}\right)^n - 1}{e^{\frac{\pi \mathrm{i}}{n}} - 1}$$

Write z in exponential form as $z = e^{\frac{\pi \mathrm{i}}{n}}$, and substitute.

$$= \frac{e^{\pi \mathrm{i}} - 1}{e^{\frac{\pi \mathrm{i}}{n}} - 1}$$

$$= \frac{-2}{e^{\frac{\pi \mathrm{i}}{n}} - 1}$$

$\left(e^{\frac{\pi \mathrm{i}}{n}}\right)^n = e^{\pi \mathrm{i}} = -1$

$$= \frac{-2e^{-\frac{\pi \mathrm{i}}{2n}}}{e^{\frac{\pi \mathrm{i}}{2n}} - e^{-\frac{\pi \mathrm{i}}{2n}}}$$

$$= \frac{-2e^{-\frac{\pi \mathrm{i}}{2n}}}{2\mathrm{i}\sin\frac{\pi}{2n}}$$

$$= \frac{\mathrm{i}e^{-\frac{\pi \mathrm{i}}{2n}}}{\sin\frac{\pi}{2n}}$$

$\frac{-2}{2\mathrm{i}} = \mathrm{i}$

$$= \frac{\mathrm{i}\left(\cos\left(-\frac{\pi}{2n}\right) + \mathrm{i}\sin\left(-\frac{\pi}{2n}\right)\right)}{\sin\frac{\pi}{2n}}$$

$$= \frac{\mathrm{i}\left(\cos\left(\frac{\pi}{2n}\right) - \mathrm{i}\sin\left(\frac{\pi}{2n}\right)\right)}{\sin\frac{\pi}{2n}}$$

$e^{-\mathrm{i}\theta} = \cos(-\theta) + \mathrm{i}\sin(-\theta)$
$= \cos\theta - \mathrm{i}\sin\theta.$

$$= \frac{\sin\left(\frac{\pi}{2n}\right) + \mathrm{i}\cos\left(\frac{\pi}{2n}\right)}{\sin\frac{\pi}{2n}}$$

Problem-solving

You know that $\sin n\theta = \frac{1}{2\mathrm{i}}(e^{\mathrm{i}n\theta} - e^{-\mathrm{i}n\theta})$.

You can use this result to simplify an expression like $e^{\mathrm{i}\theta} - 1$ by writing it in the form $e^{\frac{\mathrm{i}\theta}{2}}\left(e^{\frac{\mathrm{i}\theta}{2}} - e^{-\frac{\mathrm{i}\theta}{2}}\right) = e^{\frac{\mathrm{i}\theta}{2}}\left(2\mathrm{i}\sin\frac{\theta}{2}\right)$. In this case this is equivalent to multiplying the top and bottom of the fraction by $e^{-\mathrm{i}\frac{\pi}{2n}}$.

$$= \frac{\sin\left(\frac{\pi}{2n}\right)}{\sin\left(\frac{\pi}{2n}\right)} + \frac{i\cos\left(\frac{\pi}{2n}\right)}{\sin\left(\frac{\pi}{2n}\right)}$$

$$= 1 + i\cot\left(\frac{\pi}{2n}\right) \text{ as required}$$

Simplify.

The series $e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \ldots + e^{ni\theta}$ is geometric with first term $e^{i\theta}$, common ratio $e^{i\theta}$ and n terms.

The sum of this series is given by $S_n = \frac{e^{i\theta}(e^{ni\theta} - 1)}{e^{i\theta} - 1}$

Converting the exponential form into modulus–argument form lets you consider the real and imaginary parts of the series separately.

$e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \ldots + e^{ni\theta}$

$= (\cos\theta + i\sin\theta) + (\cos 2\theta + i\sin 2\theta) + (\cos 3\theta + i\sin 3\theta) + \ldots + (\cos n\theta + i\sin n\theta)$

$= (\cos\theta + \cos 2\theta + \cos 3\theta + \ldots + \cos n\theta) + i(\sin\theta + \sin 2\theta + \sin 3\theta + \ldots + \sin n\theta)$

Therefore,

$$\cos\theta + \cos 2\theta + \cos 3\theta + \ldots + \cos n\theta = \text{Re}\left(\frac{e^{i\theta}(e^{ni\theta} - 1)}{e^{i\theta} - 1}\right)$$

$$\sin\theta + \sin 2\theta + \sin 3\theta + \ldots + \sin n\theta = \text{Im}\left(\frac{e^{i\theta}(e^{ni\theta} - 1)}{e^{i\theta} - 1}\right)$$

Example 14

$S = e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \ldots + e^{8i\theta}$, for $\theta \neq 2n\pi$, where n is an integer.

a Show that $S = \frac{e^{\frac{9i\theta}{2}} \sin 4\theta}{\sin\frac{\theta}{2}}$

Let $P = \cos\theta + \cos 2\theta + \cos 3\theta + \ldots + \cos 8\theta$ and $Q = \sin\theta + \sin 2\theta + \sin 3\theta + \ldots + \sin 8\theta$

b Use your answer to part **a** to show that $P = \cos\frac{9\theta}{2}\sin 4\theta \operatorname{cosec}\frac{\theta}{2}$ and find similar expressions for Q and $\frac{Q}{P}$

a $e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \ldots + e^{8i\theta} = \frac{e^{i\theta}((e^{i\theta})^8 - 1)}{e^{i\theta} - 1}$

$= \frac{e^{i\theta}(e^{8i\theta} - 1)}{e^{i\theta} - 1}$

$= \frac{e^{\frac{i\theta}{2}}(e^{8i\theta} - 1)}{e^{\frac{i\theta}{2}} - e^{-\frac{i\theta}{2}}}$

S is the sum of a geometric series.
Use $\sum_{r=0}^{n-1} wz^r = \frac{w(z^n - 1)}{z - 1}$ with $w = z = e^{i\theta}$ and $n = 8$.

$(e^{i\theta})^8 = e^{8i\theta}$

Multiply the numerator and denominator by $e^{-\frac{i\theta}{2}}$

$$= \frac{e^{\frac{i\theta}{2}}(e^{8i\theta} - 1)}{2i\sin\frac{\theta}{2}}$$

$e^{\frac{i\theta}{2}} - e^{-\frac{i\theta}{2}} = 2i\sin\frac{\theta}{2}$

$$= \frac{e^{\frac{i\theta}{2}}e^{4i\theta}(e^{4i\theta} - e^{-4i\theta})}{2i\sin\frac{\theta}{2}}$$

Use the relationship $e^{8i\theta} - 1 = e^{4i\theta}(e^{4i\theta} - e^{-4i\theta})$ to rewrite the numerator.

$$= \frac{e^{\frac{9i\theta}{2}}(2i\sin 4\theta)}{2i\sin\frac{\theta}{2}}$$

$e^{4i\theta} - e^{-4i\theta} = 2i\sin 4\theta$

$$= \frac{e^{\frac{9i\theta}{2}}\sin 4\theta}{\sin\frac{\theta}{2}}$$

Simplify.

b $P = \text{Re}(S) = \text{Re}\left(\dfrac{e^{\frac{9i\theta}{2}}\sin 4\theta}{\sin\frac{\theta}{2}}\right)$

$$= \frac{\cos\frac{9\theta}{2}\sin 4\theta}{\sin\frac{\theta}{2}}$$

$$= \cos\frac{9\theta}{2}\sin 4\theta\,\text{cosec}\frac{\theta}{2}$$

$Q = \text{Im}(S) = \text{Im}\left(\dfrac{e^{\frac{9i\theta}{2}}\sin 4\theta}{\sin\frac{\theta}{2}}\right)$

$$= \frac{\sin\frac{9\theta}{2}\sin 4\theta}{\sin\frac{\theta}{2}}$$

$$= \sin\frac{9\theta}{2}\sin 4\theta\,\text{cosec}\frac{\theta}{2}$$

$$\frac{Q}{P} = \frac{\sin\frac{9\theta}{2}\sin 4\theta\,\text{cosec}\frac{\theta}{2}}{\cos\frac{9\theta}{2}\sin 4\theta\,\text{cosec}\frac{\theta}{2}} = \frac{\sin\frac{9\theta}{2}}{\cos\frac{9\theta}{2}} = \tan\frac{9\theta}{2}$$

Problem-solving

By writing each term of S in modulus–argument form you can see that P is the real part of S and Q is the imaginary part of S.

Exercise 1E

(P) **1** Given $z = e^{\frac{\pi i}{n}}$, where n is a positive integer, show that:

a $1 + z + z^2 + \ldots + z^{2n-1} = 0$ **b** $1 + z + z^2 + \ldots + z^n = i\cot\left(\dfrac{\pi}{2n}\right)$

(P) **2** Show that if $z = e^{\frac{\pi i}{2}}$, then $\sum_{r=0}^{12} z^r = 1$.

(P) **3** Show that $\sum_{r=0}^{7}(1 + i)^r = -15i$.

E/P **4** The convergent infinite series C and S are defined as

$$C = 1 + \tfrac{1}{3}\cos\theta + \tfrac{1}{9}\cos 2\theta + \tfrac{1}{27}\cos 3\theta + \ldots$$

$$S = \tfrac{1}{3}\sin\theta + \tfrac{1}{9}\sin 2\theta + \tfrac{1}{27}\sin 3\theta + \ldots$$

Hint The sum of an infinite geometric series with first term a and common ratio r is $S_\infty = \dfrac{a}{1-r}$

← Pure Year 2, Chapter 3

a Show that $C + \mathrm{i}S = \dfrac{3}{3 - \mathrm{e}^{\mathrm{i}\theta}}$ **(4 marks)**

b Hence show that $C = \dfrac{9 - 3\cos\theta}{10 - 6\cos\theta}$, and find a similar expression for S. **(4 marks)**

E/P **5** The series P and Q are defined for $0 < \theta < \pi$ as

$$P = 1 + \cos\theta + \cos 2\theta + \cos 3\theta + \ldots + \cos 12\theta$$

$$Q = \sin\theta + \sin 2\theta + \sin 3\theta + \ldots + \sin 12\theta$$

a Show that $P + \mathrm{i}Q = \dfrac{\mathrm{e}^{6\mathrm{i}\theta}\left(\mathrm{e}^{\frac{13\mathrm{i}\theta}{2}} - \mathrm{e}^{-\frac{13\mathrm{i}\theta}{2}}\right)}{\mathrm{e}^{\frac{\mathrm{i}\theta}{2}} - \mathrm{e}^{-\frac{\mathrm{i}\theta}{2}}}$ **(4 marks)**

b Deduce that $Q = \sin 6\theta \sin\dfrac{13\theta}{2}\operatorname{cosec}\dfrac{\theta}{2}$ and write down the corresponding expression for P.

You can assume the results $\sin\theta = \dfrac{\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}}{2\mathrm{i}}$ and $\cos\theta = \dfrac{\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}}{2}$ **(4 marks)**

c Hence find the values of θ, in the range $0 < \theta < \pi$, for which $P + \mathrm{i}Q$ is real. **(2 marks)**

E/P **6** Series C and S are defined as

$$C = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \binom{n}{3}\cos 3\theta + \ldots + \binom{n}{n}\cos n\theta$$

$$S = \binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \binom{n}{3}\sin 3\theta + \ldots + \binom{n}{n}\sin n\theta$$

a Show that $C = \left(2\cos\dfrac{\theta}{2}\right)^n \cos\dfrac{n\theta}{2}$ **(4 marks)**

b Show that $\dfrac{S}{C} = \tan\dfrac{n\theta}{2}$ **(3 marks)**

E/P **7** **a** Show that $(2 + \mathrm{e}^{\mathrm{i}\theta})(2 + \mathrm{e}^{-\mathrm{i}\theta}) = 5 + 4\cos\theta$. **(2 marks)**

The convergent infinite series C and S are defined by

$$C = 1 - \tfrac{1}{2}\cos\theta + \tfrac{1}{4}\cos 2\theta - \tfrac{1}{8}\cos 3\theta + \ldots$$

$$S = \tfrac{1}{2}\sin\theta - \tfrac{1}{4}\sin 2\theta + \tfrac{1}{8}\sin 3\theta + \ldots$$

b By considering $C - \mathrm{i}S$, show that $C = \dfrac{4 + 2\cos\theta}{5 + 4\cos\theta}$ and write down the corresponding expression for S. **(4 marks)**

1.6 nth roots of a complex number

You can use de Moivre's theorem to solve an equation of the form $z^n = w$, where $z, w \in \mathbb{C}$.
This is equivalent to finding the nth roots of w.

Just as a real number, x, has two square roots, $\sqrt{x}$ and $-\sqrt{x}$, any complex number has n distinct nth roots.

- **If z and w are non-zero complex numbers and n is a positive integer, then the equation $z^n = w$ has n distinct solutions.**

You can find the solutions to $z^n = w$ using de Moivre's theorem, and by considering the fact that the argument of a complex number is not unique.

Note $\cos(\theta + 2k\pi) = \cos\theta$ and $\sin(\theta + 2k\pi) = \sin\theta$ for integer values of k.

- **For any complex number $z = r(\cos\theta + \mathrm{i}\sin\theta)$, you can write $z = r(\cos(\theta + 2k\pi) + \mathrm{i}\sin(\theta + 2k\pi))$, where k is any integer.**

Example 15

a Solve the equation $z^3 = 1$.

b Represent your solutions to part **a** on an Argand diagram.

c Show that the three cube roots of 1 can be written as 1, ω and ω^2 where $1 + \omega + \omega^2 = 0$.

a $z^3 = 1$

$z^3 = \cos 0 + \mathrm{i}\sin 0$ — Start by writing 1 in modulus–argument form.

$(r(\cos\theta + \mathrm{i}\sin\theta))^3 = \cos(0 + 2k\pi) + \mathrm{i}\sin(0 + 2k\pi),\ k \in \mathbb{Z}$ — Write z in modulus–argument form, and write the general form of the argument on the right-hand side by adding integer multiples of 2π.

$r^3(\cos 3\theta + \mathrm{i}\sin 3\theta) = \cos(0 + 2k\pi) + \mathrm{i}\sin(0 + 2k\pi),\ k \in \mathbb{Z}$ — Apply de Moivre's theorem to the left-hand side of the equation.

So $r = 1$ — Compare the modulus on both sides to get $r = 1$.

$3\theta = 2k\pi$ — Compare the arguments on both sides.

$k = 0 \Rightarrow \theta = 0$, so $z_1 = \cos 0 + \mathrm{i}\sin 0 = 1$

$k = 1 \Rightarrow \theta = \frac{2\pi}{3}$

so $z_2 = \cos\left(\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + \mathrm{i}\frac{\sqrt{3}}{2}$

$k = -1 \Rightarrow \theta = -\frac{2\pi}{3}$

so $z_3 = \cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right) = -\frac{1}{2} - \mathrm{i}\frac{\sqrt{3}}{2}$

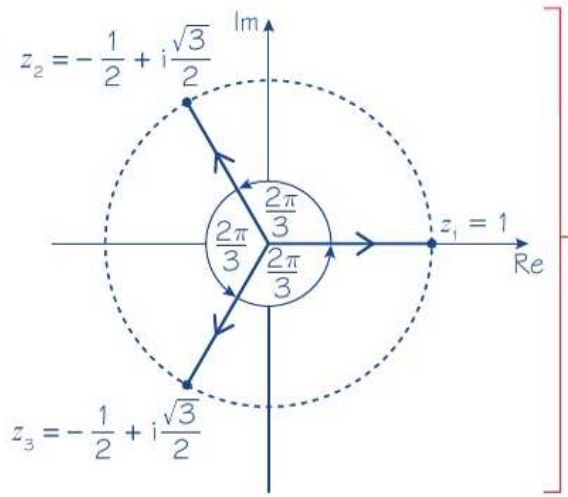
Therefore,

$z = 1,\ z = -\frac{1}{2} + \mathrm{i}\frac{\sqrt{3}}{2}$ or $z = -\frac{1}{2} - \mathrm{i}\frac{\sqrt{3}}{2}$ — These are the cube roots of unity.

Problem-solving

Choose values of k to find the three distinct roots. By choosing values on either side of $k = 0$ you can find three different arguments in the interval $[-\pi, \pi]$.

b



Plot the points $z_1 = 1$, $z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $z_3 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ on an Argand diagram:

The points z_1, z_2 and z_3 lie on a circle of radius 1 unit.

The angles between each of the vectors z_1, z_2 and z_3 are $\frac{2\pi}{3}$, as shown on the Argand diagram.

c Let $\omega = z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2} = e^{\frac{2\pi i}{3}}$

Then, $\omega^2 = \left(e^{\frac{2\pi i}{3}}\right)^2 = e^{\frac{4\pi i}{3}}$

$= e^{-\frac{2\pi i}{3}} = -\frac{1}{2} - i\frac{\sqrt{3}}{2} = z_3$

Notice that $\omega^* = \omega^2$.

$1 + \omega + \omega^2 =$

$1 + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 0$

Note It can be proved that the sum of the nth roots of unity is zero, for any positive integer $n \geqslant 2$.

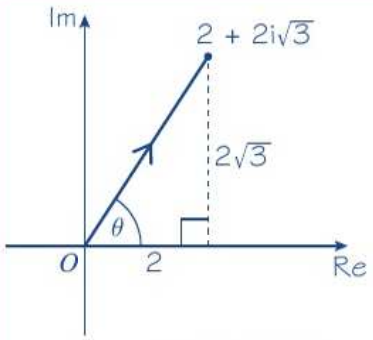
- **In general, the solutions to $z^n = 1$ are $z = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right) = e^{\frac{2\pi i k}{n}}$ for $k = 1, 2, \ldots, n$ and are known as the nth roots of unity.**

  If n is a positive integer, then there is an nth root of unity $\omega = e^{\frac{2\pi i}{n}}$ such that:

  - **the nth roots of unity are $1, \omega, \omega^2, \cdots, \omega^{n-1}$**
  - **$1, \omega, \omega^2, \cdots, \omega^{n-1}$ form the vertices of a regular n-gon**
  - **$1 + \omega + \omega^2 + \ldots + \omega^{n-1} = 0$**

Example 16

Solve the equation $z^4 = 2 + 2i\sqrt{3}$.



modulus $= \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4$

argument $= \arctan\left(\frac{2\sqrt{3}}{2}\right) = \frac{\pi}{3}$

So $z^4 = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

To solve an equation of the form $z^n = w$, start by writing w in modulus–argument form.

$(r(\cos\theta + i\sin\theta))^4$

$= 4\left(\cos\left(\frac{\pi}{3} + 2k\pi\right) + i\sin\left(\frac{\pi}{3} + 2k\pi\right)\right), k \in \mathbb{Z}$

Now let $z = r(\cos\theta + i\sin\theta)$, and write the general form of the argument on the RHS by adding integer multiples of 2π.

$r^4(\cos 4\theta + i\sin 4\theta)$

$= 4\left(\cos\left(\frac{\pi}{3} + 2k\pi\right) + i\sin\left(\frac{\pi}{3} + 2k\pi\right)\right), k \in \mathbb{Z}$

Apply de Moivre's theorem to the LHS.

So $r^4 = 4 \Rightarrow r = \sqrt[4]{4} = \sqrt{2}$

Compare the modulus on both sides to get $r = \sqrt{2}$.

$4\theta = \frac{\pi}{3} + 2k\pi$

Compare the arguments on both sides.

$k = 0 \Rightarrow \theta = \frac{\pi}{12}$, so $z_1 = \sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

$k = 1 \Rightarrow \theta = \frac{7\pi}{12}$, so $z_2 = \sqrt{2}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$

When $k = 1$, $4\theta = \frac{\pi}{3} + 2\pi$

$\Rightarrow \theta = \frac{\pi}{12} + \frac{2\pi}{4} = \frac{7\pi}{12}$

$k = -1 \Rightarrow \theta = -\frac{5\pi}{12}$, so $z_3 = \sqrt{2}\left(\cos\left(-\frac{5\pi}{12}\right) + i\sin\left(-\frac{5\pi}{12}\right)\right)$

$k = -2 \Rightarrow \theta = -\frac{11\pi}{12}$, so $z_4 = \sqrt{2}\left(\cos\left(-\frac{11\pi}{12}\right) + i\sin\left(-\frac{11\pi}{12}\right)\right)$

Watch out Make sure you choose n **consecutive** values of k to get n distinct roots. If an argument is not in the interval $[-\pi, \pi]$ you can add or subtract a multiple of 2π.

or $z = \sqrt{2}e^{\frac{\pi i}{12}}$, $z = \sqrt{2}e^{\frac{7\pi i}{12}}$, $z = \sqrt{2}e^{-\frac{5\pi i}{12}}$ or $z = \sqrt{2}e^{-\frac{11\pi i}{12}}$

These are the solutions in the form $re^{i\theta}$.

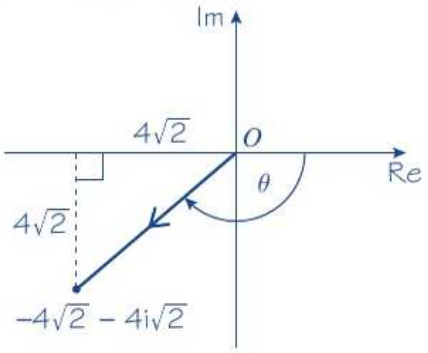
You can also use the exponential form of a complete number when solving equations.

Example 17

Solve the equation $z^3 + 4\sqrt{2} + 4i\sqrt{2} = 0$.

$z^3 + 4\sqrt{2} + 4i\sqrt{2} = 0$

$z^3 = -4\sqrt{2} - 4i\sqrt{2}$



Find the modulus and argument of $-4\sqrt{2} - 4i\sqrt{2}$.

$\text{modulus} = \sqrt{(-4\sqrt{2})^2 + (-4\sqrt{2})^2} = \sqrt{32 + 32} = \sqrt{64} = 8$

$\text{argument} = -\pi + \arctan\left(\frac{4\sqrt{2}}{4\sqrt{2}}\right) = -\pi + \frac{\pi}{4} = -\frac{3\pi}{4}$

Write $z = re^{i\theta}$ and use $(re^{i\theta})n = r^n e^{in\theta}$. Remember to write the general form of the argument on the right-hand side by adding integer multiples of 2π.

$(re^{i\theta})^3 = 8e^{\left(-\frac{3\pi}{4} + 2k\pi\right)i}$

$r^3 e^{3i\theta} = 8e^{\left(-\frac{3\pi}{4} + 2k\pi\right)i}$

Compare the modulus on both sides to get $r = 2$.

So $r^3 = 8 \Rightarrow r = \sqrt[3]{8} = 2$

Compare the arguments on both sides.

$3\theta = -\frac{3\pi}{4} + 2k\pi$

Choose values of k to find three distinct roots. Either choose values that produce arguments in the interval $-\pi < \theta \leqslant \pi$, or add or subtract multiples of 2π as necessary.

$k = 0 \Rightarrow \theta = -\frac{\pi}{4}$, so $z_1 = 2e^{\frac{-\pi i}{4}}$

$k = 1 \Rightarrow \theta = \frac{5\pi}{12}$, so $z_2 = 2e^{\frac{5\pi i}{12}}$

$k = -1 \Rightarrow \theta = -\frac{11\pi}{12}$, so $z_3 = 2e^{\frac{-11\pi i}{12}}$

or $z = 2\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$, $z = 2\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$

or $z = 2\left(\cos\left(-\frac{11\pi}{12}\right) + i\sin\left(-\frac{11\pi}{12}\right)\right)$.

Exercise 1F

1 Solve the following equations, expressing your answers for z in the form $x + iy$, where $x, y \in \mathbb{R}$.

a $z^4 - 1 = 0$ **b** $z^3 - i = 0$ **c** $z^3 = 27$

d $z^4 + 64 = 0$ **e** $z^4 + 4 = 0$ **f** $z^3 + 8i = 0$

2 Solve the following equations, expressing the roots in the form $r(\cos\theta + i\sin\theta)$, where $-\pi < \theta \leqslant \pi$.

a $z^7 = 1$ **b** $z^4 + 16i = 0$ **c** $z^5 + 32 = 0$

d $z^3 = 2 + 2i$ **e** $z^4 + 2i\sqrt{3} = 2$ **f** $z^3 + 32\sqrt{3} + 32i = 0$

3 Solve the following equations, expressing the roots in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leqslant \pi$. Give θ to 2 decimal places.

a $z^4 = 3 + 4i$ **b** $z^3 = \sqrt{11} - 4i$ **c** $z^4 = -\sqrt{7} + 3i$

(P) **4** **a** Find the three roots of the equation $(z + 1)^3 = -1$.
Give your answers in the form $x + iy$, where $x, y \in \mathbb{R}$.

b Plot the points representing these three roots on an Argand diagram.

c Given that these three points lie on a circle, find its centre and radius.

(P) **5** **a** Find the five roots of the equation $z^5 - 1 = 0$.
Give your answers in the form $r(\cos\theta + i\sin\theta)$, where $-\pi < \theta \leqslant \pi$.

b Hence or otherwise, show that

$$\cos\left(\frac{2\pi}{5}\right) + \cos\left(\frac{4\pi}{5}\right) = -\frac{1}{2}$$

Problem-solving

Use the fact that the sum of the five roots of unity is zero.

(E) **6** **a** Find the modulus and argument of $-2 - 2i\sqrt{3}$. **(2 marks)**

b Hence find all the solutions of the equation $z^4 + 2 + 2i\sqrt{3} = 0$.
Give your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leqslant \pi$ and illustrate the roots on an Argand diagram. **(4 marks)**

(E) **7** Find the four distinct roots of the equation $z^4 = 2(1 - i\sqrt{3})$ in exponential form, and show these roots on an Argand diagram. **(7 marks)**

(E/P) **8** $z = \sqrt{6} + i\sqrt{2}$

a Find the modulus and argument of z. **(2 marks)**

b Find the values of w such that $w^3 = z^4$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leqslant \pi$. **(4 marks)**

(P) **9 a** Solve the equation

$$1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 = 0$$

b Hence deduce that $(z^2 + 1)$ and $(z^4 + 1)$ are factors of $1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7$.

Problem-solving

$1 + z + z^2 + z^3 + \ldots + z^7$ is the sum of a geometric series.

Challenge

a Find the six roots of the equation $z^6 = 1$ in the form $e^{i\theta}$, where $-\pi < \theta \leqslant \pi$.

b Hence show that the solutions to $(z + 1)^6 = z^6$ are $z = -\frac{1}{2} + \frac{1}{2}i\cot\left(\frac{k\pi}{6}\right)$, $k = 1, 2, 3, 4, 5$.

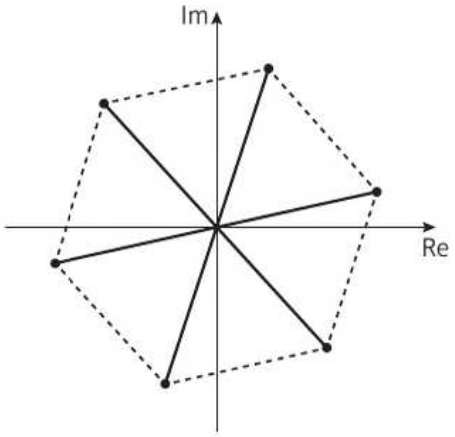
1.7 Solving geometric problems

You can use properties of complex nth roots to solve geometric problems.

- **The nth roots of any complex number a lie at the vertices of a regular n-gon with its centre at the origin.**

Notation The centre of a regular polygon is considered to be the centre of the circle that passes through all of its vertices.

The orientation and size of the regular polygon will depend on a.



For example, the sixth roots of 7 + 24i form this regular hexagon. Each vertex of the hexagon is equidistant from the origin, which lies at the centre of the circle passing through all six vertices.

Online Explore nth roots of complex numbers in an Argand diagram using GeoGebra.



You can find the vertices of this regular polygon by finding a single vertex, and rotating that point around the origin. This is equivalent to multiplying by the nth roots of unity.

- **If z_1 is one root of the equation $z^n = s$, and 1, ω, ω^2, …, ω^{n-1} are the nth roots of unity, then the roots of $z^n = s$ are given by z_1, $z_1\omega$, $z_1\omega^2$, …, $z_1\omega^{n-1}$.**

Example 18

The point $P(\sqrt{3}, 1)$ lies at one vertex of an equilateral triangle. The centre of the triangle is at the origin.

a Find the coordinates of the other vertices of the triangle.

b Find the area of the triangle.

Problem-solving

Consider the Cartesian coordinate plane as an Argand diagram. The vertices of the triangle will correspond to the cube roots of $(\sqrt{3} + i)^3$. You can find these roots by multiplying $\sqrt{3} + i$ by the cube roots of unity.

a The cube roots of unity are $1, \omega, \omega^2$ where $\omega = e^{\frac{2\pi i}{3}}$.

Use $\omega = e^{\frac{2\pi i}{n}}$ to write down the cube root of unity.

$\sqrt{3} + i = 2e^{\frac{\pi i}{6}}$

$|\sqrt{3} + i| = 2$ and $\arg(\sqrt{3} + i) = \frac{\pi}{6}$

So the vertices are at:

$2e^{\frac{\pi i}{6}} = \sqrt{3} + i$

$2e^{\frac{\pi i}{6}}\left(e^{\frac{2\pi i}{3}}\right) = 2e^{\frac{5\pi i}{6}} = -\sqrt{3} + i$

$2e^{\frac{\pi i}{6}}\left(e^{\frac{2\pi i}{3}}\right)^2 = 2e^{\frac{\pi i}{6}}\left(e^{\frac{4\pi i}{3}}\right) = 2e^{\frac{9\pi i}{6}} = 2e^{-\frac{\pi i}{2}} = -2i$

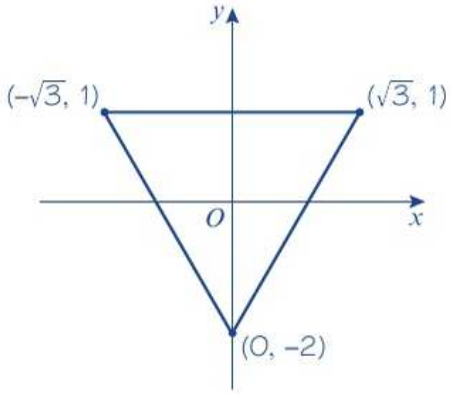
You know that one cube root of $(\sqrt{3} + i)^3$ is $\sqrt{3} + i$. Multiply this by the cube roots of unity to find the other roots.

So the coordinates of the vertices of the triangle are

$(\sqrt{3}, 1)$, $(-\sqrt{3}, 1)$ and $(0, -2)$.

Write your answers as Cartesian coordinates.

b



Area $= \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times 2\sqrt{3} \times 3$

$= 3\sqrt{3}$

Exercise 1G

(P) **1** Find the coordinates of the vertices of the following regular polygons with centres at the origin.

a Equilateral triangle with one vertex at $(0, 4)$

b Square with one vertex at $(5, 0)$

c Regular pentagon with one vertex at $(-1, \sqrt{3})$

d Regular hexagon with one vertex at $(2, 2)$

(P) **2** Find the coordinates of the vertices of an equilateral triangle with centre $(2, 3)$ and one vertex at $(3, -2)$.

E/P **3** The triangle OAB in an Argand diagram is equilateral. O is the origin and A corresponds to the complex number $\sqrt{3}(1 - \mathrm{i})$. B is represented by the complex number b.

Find the two possibilities for b in the form $re^{i\theta}$. Illustrate the two possibilities for OAB in a sketch. **(5 marks)**

E/P **4 a** Find the 4th roots of $-12\mathrm{i}$ in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leqslant \pi$. Illustrate these roots on an Argand diagram. **(6 marks)**

Let the points representing these roots on an Argand diagram, taken in order of increasing θ, be A, B, C, D. The midpoints of the sides of $ABCD$ represent the 4th roots of a complex number w.

b Find w. **(4 marks)**

E/P **5** P is one vertex of a regular hexagon in an Argand diagram. The centre of the hexagon is at the origin. P corresponds to the complex number $8 + 8\mathrm{i}$.

a Find, in the form $a + b\mathrm{i}$, the complex numbers corresponding to the other vertices of the hexagon, and illustrate these on an Argand diagram. **(5 marks)**

b The six complex numbers corresponding to the vertices of the hexagon are squared to form the vertices of a new figure. Find, in the form $a + b\mathrm{i}$, the complex numbers corresponding to the other vertices of the new figure. Find the area of the new figure. **(4 marks)**

E/P **6** An ant walks forward one unit and then turns to the right by $\frac{2\pi}{9}$. It repeats this a further three times. Show that the distance of the ant from its initial position is $\dfrac{\sin\left(\frac{4\pi}{9}\right)}{\sin\left(\frac{\pi}{9}\right)}$ **(6 marks)**

Mixed exercise 1

P **1 a** Use $e^{i\theta} = \cos\theta + \mathrm{i}\sin\theta$ to show that $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$.

b Hence prove that $\cos A \cos B \equiv \dfrac{\cos(A + B) + \cos(A - B)}{2}$

E/P **2** Given that $z = r(\cos\theta + \mathrm{i}\sin\theta)$, $r \in \mathbb{R}$, prove by induction that $z^n = r^n(\cos n\theta + \mathrm{i}\sin n\theta)$, $n \in \mathbb{Z}^+$. **(5 marks)**

3 Express $\dfrac{(\cos 3x + \mathrm{i}\sin 3x)^2}{\cos x - \mathrm{i}\sin x}$ in the form $\cos nx + \mathrm{i}\sin nx$ where n is an integer to be determined.

4 Use de Moivre's theorem to evaluate:

a $(-1 + \mathrm{i})^8$ **b** $\dfrac{1}{\left(\frac{1}{2} - \frac{1}{2}\mathrm{i}\right)^{16}}$

E/P **5 a** Given $z = \cos\theta + \mathrm{i}\sin\theta$, use de Moivre's theorem to show that $z^n + \dfrac{1}{z^n} = 2\cos n\theta$. **(4 marks)**

b Express $\left(z^2 + \dfrac{1}{z^2}\right)^3$ in terms of $\cos 6\theta$ and $\cos 2\theta$. **(3 marks)**

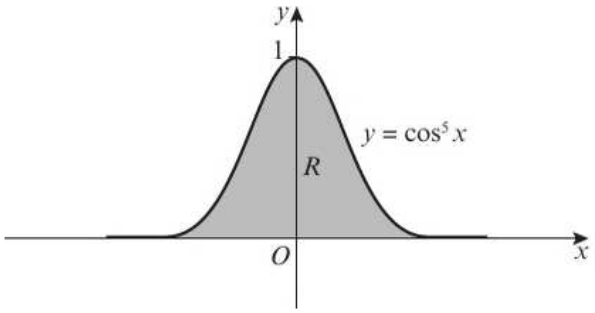
c Hence, or otherwise, find constants a and b such that $\cos^3 2\theta = a\cos 6\theta + b\cos 2\theta$. **(3 marks)**

d Hence, or otherwise, show that $\displaystyle\int_0^{\frac{\pi}{6}} \cos^3 2\theta \, \mathrm{d}\theta = k\sqrt{3}$, where k is a rational constant. **(4 marks)**

E/P **6 a** Show that

$$\cos^5\theta \equiv \frac{1}{16}(\cos 5\theta + 5\cos 3\theta + 10\cos\theta)$$ **(5 marks)**

The diagram shows the curve with equation $y = \cos^5 x$, $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$. The finite region R is bounded by the curve and the x-axis.



b Calculate the exact area of R. **(6 marks)**

E/P **7 a** Show that

$$\sin^6\theta \equiv -\frac{1}{32}(\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10)$$ **(5 marks)**

b Using the substitution $\alpha = \left(\frac{\pi}{2} - \theta\right)$, or otherwise, find a similar identity for $\cos^6\theta$. **(3 marks)**

c Given that $\int_0^a \cos^6\theta + \sin^6\theta\, d\theta = \frac{5\pi}{32}$, find the exact value of a. **(5 marks)**

E/P **8** Use de Moivre's theorem to show that

$$\sin 6\theta \equiv \sin 2\theta(16\cos^4\theta - 16\cos^2\theta + 3)$$ **(5 marks)**

E/P **9 a** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **(5 marks)**

b Hence find all solutions to the equation

$$16x^5 - 20x^3 + 5x + 1 = 0$$

giving your answers to 3 decimal places where necessary. **(5 marks)**

E/P **10 a** Show that

$$\sin^5\theta \equiv \frac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$$ **(5 marks)**

b Hence solve the equation

$$\sin 5\theta - 5\sin 3\theta + 9\sin\theta = 0 \text{ for } 0 \leqslant \theta < \pi$$ **(4 marks)**

E/P **11 a** Use de Moivre's theorem to show that $\cos 5\theta \equiv \cos\theta(16\cos^4\theta - 20\cos^2\theta + 5)$ **(5 marks)**

b By solving the equation $\cos 5\theta = 0$, deduce that $\cos^2\left(\frac{\pi}{10}\right) = \frac{5+\sqrt{5}}{8}$ **(4 marks)**

c Hence, or otherwise, write down the exact values of $\cos^2\left(\frac{3\pi}{10}\right)$, $\cos^2\left(\frac{7\pi}{10}\right)$ and $\cos^2\left(\frac{9\pi}{10}\right)$. **(3 marks)**

E/P **12** **a** Use de Moivre's theorem to find an expression for $\tan 3\theta$ in terms of $\tan\theta$. **(4 marks)**

b Deduce that $\cot 3\theta = \dfrac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}$ **(2 marks)**

E/P **13** The infinite series C and S are defined as

$$C = 1 + k\cos\theta + k^2\cos 2\theta + k^3\cos 3\theta + \ldots$$

$$S = k\sin\theta + k^2\sin 2\theta + k^3\sin 3\theta + \ldots$$

where k is a real number and $|k| < 1$.

By considering $C + \mathrm{i}S$, show that $C = \dfrac{1 - k\cos\theta}{1 + k^2 - 2k\cos\theta}$ and write down the corresponding expression for S. **(8 marks)**

E **14** **a** Express $4 - 4\mathrm{i}$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $r > 0$, $-\pi < \theta \leqslant \pi$, where r and θ are exact values. **(2 marks)**

b Hence, or otherwise, solve the equation $z^5 = 4 - 4\mathrm{i}$, leaving your answers in the form $z = R\mathrm{e}^{\mathrm{i}k\pi}$, where R is the modulus of z and k is a rational number such that $-1 \leqslant k \leqslant 1$. **(4 marks)**

c Show on an Argand diagram the points representing the roots. **(2 marks)**

E/P **15** **a** Find the cube roots of $2 - 2\mathrm{i}$ in the form $r\mathrm{e}^{\mathrm{i}\theta}$ where $r > 0$ and $-\pi < \theta \leqslant \pi$. **(5 marks)**

These cube roots are represented by points A, B and C in the Argand diagram, with A in the fourth quadrant and ABC going anticlockwise. The midpoint of AB is M, and M represents the complex number w.

b Draw an Argand diagram, showing the points A, B, C and M. **(2 marks)**

c Find the modulus and argument of w. **(2 marks)**

d Find w^6 in the form $a + b\mathrm{i}$. **(3 marks)**

E/P **16** An equilateral triangle has its centre at the origin and one vertex at the point (2, 1).

a Find the coordinates of the other two vertices. **(4 marks)**

b Show that the length of one side of the triangle is $\sqrt{15}$. **(2 marks)**

Challenge

Show that the points on an Argand diagram that represent the roots of $\left(\dfrac{z+1}{z}\right)^6 = 1$ lie on a straight line.